



1. Introduction

This report summarises the progress of the team for the February reporting cycle. The main achievements include migrating the BFMC simulator from ROS to ROS2, validating the Ackermann vehicle behaviour and control interface within Gazebo, and setting the first perception to control pipeline linking camera input to steering reference. Additionally, the camera and Jetson were integrated and this pipeline was tested on the real vehicle after the simulations.

2. Planned activities

At the beginning of this reporting period, several activities were planned to allow the transition from infrastructure preparation to perception driven control. The planned activities were as follows:

Simulator Migration to ROS2: Porting the BFMC simulator from ROS to ROS2 while removing external dependencies and restructuring the project into modular packages suitable for further development and testing.

Control Interface Validation: Ensuring that the simulator sticks to standard ROS control interfaces, allowing the vehicle to be driven using `/drive_cmd` command and teleoperation nodes.

Perception Pipeline in Simulation: Integrating virtual cameras in Gazebo and using their image topics to test a lane detection algorithm capable of producing a steering reference.

Physical Camera and Jetson Integration: Bringing up the stereo cameras on the Jetson platform, calibrating them, and preparing the system to run the perception pipeline on the real vehicle.

3. Status of planned activities

A. Physical Camera & Jetson Integration

Status: In Progress

Implementation: The Jetson was flashed and prepared as the primary computing unit for perception tasks. Arducam drivers were installed for the IMX708 cameras, and ROS2 launch files were written using GStreamer and NVIDIA Argus to publish both camera streams to ROS image topics. Calibration was performed for both cameras using an 8×6 checkerboard pattern to ensure accurate image geometry.

Difficulties: The Jetson does not natively support the IMX708 sensors, resulting in a severe red tint in the captured images due to incompatible ISP configurations.

Solution: ISP configuration files intended for a supported sensor (IMX477) were adapted and used in place of the default settings. This resolved the colour distortion and produced usable image streams for perception processing.

In parallel, the steering system on the vehicle was calibrated using the dashboard interface and the Nucleo firmware was reflashed to ensure accurate steering response when the perception pipeline is executed on the real car.

B. Simulator Migration & Control Interface Validation

Status: Completed

Implementation: The BFMC simulator was migrated from ROS to ROS2 and restructured into modular packages separating simulation logic and Gazebo assets. During this process, unnecessary external dependencies were removed to simplify the build process and improve maintainability. The URDF model and Ackermann steering behaviour were reviewed and corrected to ensure that the simulated vehicle accurately reflects the physical car's kinematics. Topic remapping and interface adjustments were implemented so that the vehicle could be driven using conventional ROS teleoperation nodes and manual command publishing. This allowed the simulator to share the same control interface as the real vehicle.

C. Perception Pipeline Validation in Simulation

Status: Completed

Implementation: Virtual cameras were integrated into the Gazebo environment and their image topics were verified for use in perception testing. A lane detection node was then integrated into ROS2 and connected to the camera feed.

The image processing pipeline operates by extracting a region of interest from the lower portion of the image, applying adaptive thresholding and morphological filtering, and scanning the image row-by-row to identify the left and right lane boundaries. From these boundaries, the centre of the lane is computed and averaged across multiple rows to obtain a stable estimate.

The lateral offset of the vehicle from the lane centre is then calculated in pixel space relative to the image midpoint. This offset is smoothed over time and used as a steering reference for generating angular velocity commands. Using this method, the simulator was able to convert camera input into a steering response, establishing a complete perception-to-control chain in a safe testing environment.

4. General status of the project

At its current stage, the project has progressed from infrastructure preparation to the first implementation of perception-driven control. The simulator now provides a reliable environment where camera input can be translated into steering behaviour using the same control interface as the physical vehicle.

Initial tests were carried out on the physical car, where live camera input from the Jetson was processed by the lane detection node and translated into steering behaviour through the Nucleo. While the behaviour needs further tuning, the perception-to-control chain is now functioning on hardware.

5. Upcoming activities

The next phase will focus on environment perception and localization, combining vision-based methods with map-referenced information for improved vehicle awareness. Dataset collection and preparation will support model training, testing, and validation of perception-to-control performance in simulation and on the real vehicle. Finally, end-to-end autonomy will be expanded by strengthening camera-to-steering control for lane keeping and introducing reactions to standard traffic elements such as intersections, signs, and obstacles.